LAB # 3

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Lab Section : C

1. Introduction

Resonance is an important concept in AC circuit analysis and plays a key role in many electronic systems. It occurs in circuits that contain inductors and capacitors when the inductive reactance and capacitive reactance become equal in magnitude but opposite in phase. Under this condition, the reactive effects cancel each other and the circuit behaves as if it contains only resistance. As a result, the circuit exhibits a distinct response at a specific frequency called the resonant frequency.

In RLC circuits, the impedance of the circuit depends on the frequency of the input signal. The inductive reactance increases with frequency while the capacitive reactance decreases. At low frequencies the capacitor dominates the circuit behavior, whereas at high frequencies the inductor becomes dominant. At the resonant frequency, these two reactances become equal, which causes the overall reactive component of the impedance to vanish. In a series RLC circuit, this condition leads to minimum impedance and therefore maximum current through the circuit.

The resonant frequency of an RLC circuit depends on the values of the inductance and capacitance in the circuit and can be expressed as
$f_o = 1 / (2\pi\sqrt{LC})$.
Another important characteristic of resonant circuits is the bandwidth, which represents the range of frequencies over which the circuit responds effectively. The quality factor (Q) is also used to describe how sharp the resonance is and how selectively the circuit responds to a narrow range of frequencies.

Resonant circuits are widely used in electronic and communication systems such as radio receivers, filters, oscillators, and signal processing devices. Understanding the resonance phenomenon is therefore essential for analyzing and designing many practical circuits.

The objective of this experiment is to study the behavior of RLC resonant circuits and observe how the circuit response changes with frequency. By applying sinusoidal input signals and measuring voltages across different components, the resonant frequency, bandwidth, and general frequency response of the circuit can be examined and compared with theoretical predictions.

2. Pre-Lab Work

3. Data/Results

Part 1 – Frequency Response of RC Circuits

Building Block

In this part of the experiment, a series RC circuit was constructed on a breadboard in order to investigate the frequency response of the circuit. A sinusoidal signal generated by the function generator was applied to the circuit, and both the input and output signals were monitored using an oscilloscope. Channel 1 of the oscilloscope was connected to the input signal, while Channel 2 was connected to the output voltage measured across the capacitor.

By sweeping the frequency of the input signal over a wide range, the magnitude of the output voltage and the phase difference between the input and output signals were measured. These measurements allowed the behavior of the RC circuit to be analyzed as a function of frequency and helped demonstrate the filtering properties of the circuit.

Components

- 1 kΩ resistor
- 47 nF capacitor
- Function generator
- Oscilloscope
- Breadboard and jumper wires

Experimental Setup

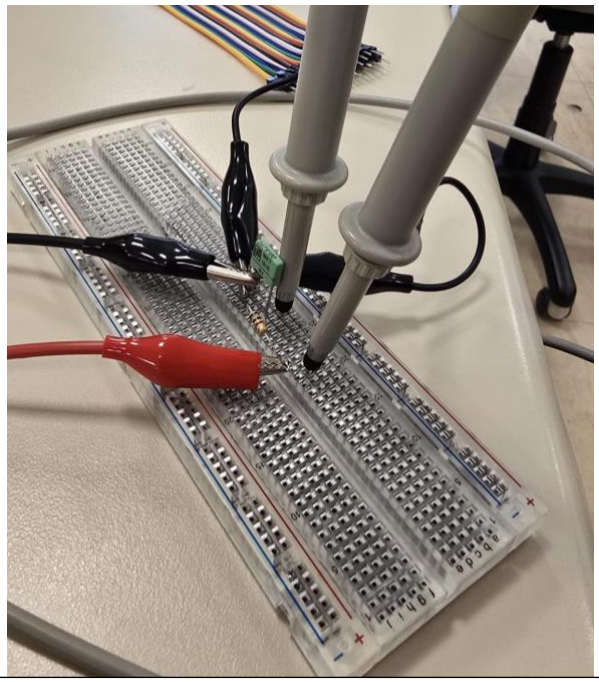


Figure 5. Breadboard implementation of the RC circuit used in the experiment.

Experimental Setup

The circuit was constructed according to the schematic shown in Figure 5. A sinusoidal input signal of 5 Vpp was generated using the function generator. The input voltage was observed through Channel 1 of the oscilloscope, while the output voltage across the capacitor was measured through Channel 2. The physical implementation of the circuit on the breadboard is shown in the experimental setup figure 5.

Theory

The behavior of a series RC circuit strongly depends on the frequency of the input signal because the impedance of the capacitor varies with frequency. The impedance of the capacitor is given by Zc = 1/(jωC)

where ω represents the angular frequency and C is the capacitance. Since the capacitive impedance is inversely proportional to frequency, its value becomes very large at low frequencies and very small at high frequencies.

At low frequencies, the capacitor behaves almost like an open circuit because its reactance is large. Under this condition, most of the input voltage appears across the capacitor, resulting in an output voltage that is approximately equal to the input voltage. As the frequency increases, the capacitive reactance decreases and the capacitor begins to allow more current to pass through the circuit. Consequently, the voltage across the capacitor decreases relative to the input voltage.

The transfer function of a series RC circuit with the output taken across the capacitor can be written as Vout / Vin = 1 / (1 + jωRC)

This transfer function shows that both the magnitude and the phase of the output signal depend on the frequency of the input signal. As frequency increases, the magnitude of the output signal decreases and the phase difference between the input and output signals increases. This behavior corresponds to the characteristics of a first-order low-pass filter.

Measurements and Observations

To obtain the frequency response of the circuit, the input signal frequency was manually swept using the function generator. For each selected frequency, the amplitude of the output voltage and the phase difference between the input and output signals were measured using the oscilloscope. The measurements were recorded for several frequencies ranging from 100 Hz to 100 kHz.

Vout

Frequency (Hz)	100	500	1000	2000	3400	10000	100K
Magnitude (V)	5.0V	5.0V	4.8V	4.20V	3.42V	1.53V	161 mV
Phase (°)	3.46°	10.66°	21.35°	34.38°	47.4	70.26°	87.91°

Vin

Frequency (Hz)	100	500	1000	2000	3400	10000	100K
Magnitude (V)	5.2V	5.2V	5.1V	5.1V	5.1V	4.9V	5V
Phase (°)	-3.46°	-10.66°	-21.35°	-34.38°	-47.4°	-70.26°	-87.91°

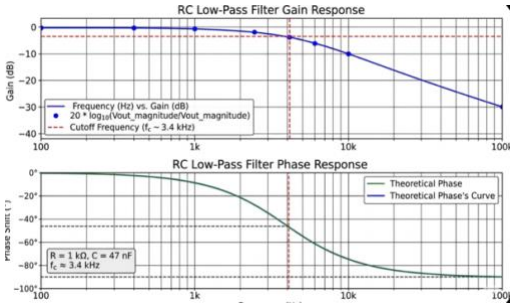


Table and graph 1. Measured frequency, magnitude, and phase values of the input (Vin) and output (Vout) signals obtained from

The results show that at low frequencies such as 100 Hz and 500 Hz, the output voltage magnitude is nearly equal to the input voltage, with measured values close to 5 V. This indicates that most of the input signal appears across the capacitor, which is consistent with the expected behavior of the circuit at low frequencies.

As the frequency increases, the magnitude of the output voltage gradually decreases. For example, the output voltage drops to approximately 3.42 V at 3400 Hz and further decreases to about 1.53 V at 10 kHz. At very high frequencies, such as 100 kHz, the output voltage becomes very small (approximately 161 mV). This reduction in output voltage occurs because the capacitive reactance decreases as frequency increases, causing a larger portion of the voltage to drop across the resistor instead of the capacitor.

The phase measurements also follow the theoretical expectations for an RC circuit. At low frequencies, the phase difference between the input and output signals is relatively small. However, as the frequency increases, the phase shift becomes larger in magnitude. The output signal increasingly lags behind the input signal, approaching a phase shift close to 90 degrees at high frequencies. The measured phase difference at 100 kHz is approximately 87.8°, which is very close to the theoretical limit.

The input voltage remained approximately constant at around 5 V for all frequency values, confirming that the changes observed in the output voltage are due to the frequency-dependent behavior of the RC circuit rather than variations in the input signal.

Figure 6 shows an example oscilloscope measurement where both the input (Channel 1) and output (Channel 2) signals can be observed. The amplitude difference and phase shift between the two signals clearly illustrate the filtering effect of the RC circuit.



Figure 6. Oscilloscope measurement showing input (CH1) and output (CH2) signals of the RC circuit for arbitrary frequencies.

Measurement of the 3-dB Frequency

In this part of the experiment, the 3-dB frequency, also called the half-power frequency or cutoff frequency, was determined for the RC circuit. For a first-order RC low-pass filter, the 3-dB point is the

frequency at which the output magnitude decreases to 1/√2 of its maximum low-frequency value. Since power is proportional to the square of voltage, a drop to 1/√2 in voltage corresponds to a reduction of one-half in power. For this reason, this point is referred to as the half-power point.

For the RC circuit used in this experiment, the theoretical cutoff frequency is given by fc = 1 / (2πRC) Using R = 1 kΩ and C = 47 nF, fc = 1 / (2π × 1000 × 47 × 10^-9), fc ≈ 3386 Hz

Therefore, the theoretical 3-dB frequency is approximately 3.39 kHz.

Experimentally, the input frequency was adjusted until the output voltage dropped to a value close to the 3-dB level. From the oscilloscope measurement, at a frequency of 3.4 kHz, the input voltage was measured as approximately 5.1 V peak-to-peak and the output voltage was measured as approximately 3.42 V peak-to-peak. The phase difference between input and output was also measured as 47.4°.

This result is very close to the expected cutoff behavior. The measured frequency of 3.4 kHz agrees well with the theoretical value of 3.386 kHz. In addition, for a first-order RC low-pass filter, the phase shift at the cutoff frequency is theoretically 45°. The measured phase shift of 47.4° is also close to this expected value, which further supports that the circuit was operating near the 3-dB point.

At the cutoff frequency, the magnitudes of the resistor voltage and capacitor voltage become comparable, and the circuit transitions from the passband region to the attenuation region. Below this frequency, the capacitor voltage remains close to the input voltage. Above this frequency, the output across the capacitor begins to decrease more rapidly as frequency increases. Therefore, the 3-dB point represents the boundary between these two operating regions and is one of the most important parameters for describing the frequency response of the circuit.

The small difference between the theoretical and measured values can be attributed to practical factors such as component tolerances, non-ideal capacitor behavior, oscilloscope reading limitations, and minor variations in the applied input voltage. Despite these factors, the measured cutoff frequency was found to be in very good agreement with theory.

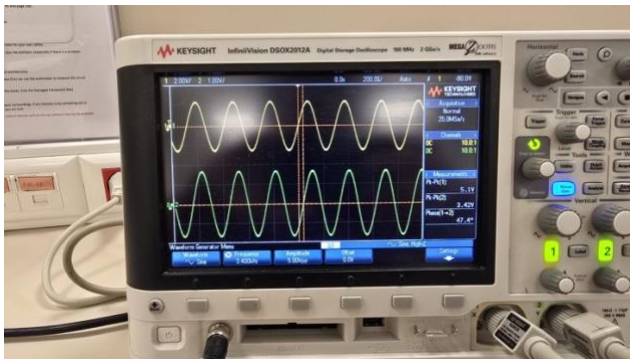


Figure 7. Oscilloscope measurement near the 3-dB frequency, showing the input and output signals at approximately 3.4 kHz.

Part 2 – Frequency Response of Series RLC Circuits

Building Block

In this part of the experiment, a series RLC circuit was constructed in order to investigate the frequency response and resonance behavior of the circuit. Unlike the RC circuit studied in Part 1, the RLC circuit contains both an inductor and a capacitor, which introduces a resonant frequency at which the inductive and capacitive reactances become equal in magnitude. At this frequency, the circuit exhibits a unique behavior that can be observed both in terms of voltage relationships and signal amplitudes.

The circuit was built using a resistor, an inductor, and a capacitor connected in series. Since a 40 Ω resistor was not directly available, this resistance value was obtained by connecting three 120 Ω resistors in parallel. The equivalent resistance of this combination is Req = 120/3 = 40 Ω

which satisfies the required resistance value for the experiment.

Components

- Equivalent 40 Ω resistor formed by three 120 Ω resistors connected in parallel
- 1 µH inductor
- 1 nF capacitor
- Function generator
- Oscilloscope
- Breadboard and jumper wires

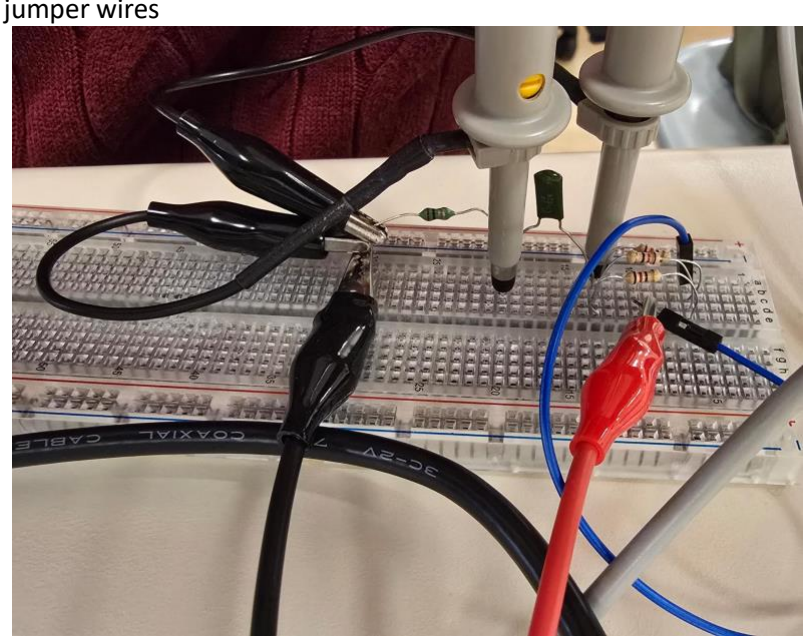


Figure 8. Breadboard implementation of the series RLC circuit used in Part 2.

Experimental Setup

The circuit was assembled according to the series RLC configuration given in the laboratory manual. The resistor was connected in series with the capacitor and inductor, and the input sinusoidal signal was applied from the function generator. The oscilloscope probes were connected as required in order to observe the voltages needed for resonance analysis.

According to the measurement method, one oscilloscope probe was used to monitor the voltage across the capacitor and inductor together, while the other probe was used to monitor the voltage across the inductor alone. Then, using the Math function of the oscilloscope with the operation Channel 1 minus Channel 2, the capacitor voltage was obtained as Vc = (Vc + VL) - VL

This method allows the capacitor voltage to be measured indirectly even though the capacitor is not directly referenced to ground. This is necessary because the oscilloscope channels share a common ground, and direct measurement across floating components in a series branch may otherwise cause incorrect connections or grounding problems.

Theory

The impedance of a series RLC circuit is given by $Z = R + j(\omega L - 1/\omega C)$

This expression shows that the total impedance depends on resistance, inductance, capacitance, and frequency. The resistor contributes only a real term, while the inductor and capacitor contribute frequency-dependent imaginary terms. The inductive reactance is $X_L = \omega L$

and the capacitive reactance is $X_C = 1/\omega C$

These two reactances vary in opposite ways with frequency. As the frequency increases, the inductive reactance increases, whereas the capacitive reactance decreases.

At low frequencies, the capacitive reactance is dominant because X_C is large, while X_L remains small. In this region, the circuit behaves mainly as a capacitive circuit. At high frequencies, the inductor becomes dominant because X_L increases and X_C becomes small. In this case, the circuit behaves mainly as an inductive circuit.

At one specific frequency, called the resonant frequency, the inductive and capacitive reactances become equal: $X_L = X_C$ or equivalently, $\omega_0 L = 1/\omega_0 C$

From this condition, the resonant angular frequency and resonant frequency are obtained as $\omega_0 = 1/\sqrt{LC}$ $f_0 = 1 / (2\pi\sqrt{LC})$

At resonance, the reactive parts cancel each other, so the total impedance becomes purely resistive: $Z = R$

This means that the impedance reaches its minimum value in a series RLC circuit, and the circuit current becomes maximum for a given input voltage. Since the same current flows through all elements in a series circuit, the voltages across the resistor, inductor, and capacitor can become significantly different even though the source voltage is fixed.

An important feature of resonance is that the voltages across the inductor and capacitor can each become large, even though they are opposite in phase and tend to cancel each other in the total loop. Ideally, at resonance, the magnitudes of VL and VC are equal, but their phases are opposite. Therefore, their sum becomes very small: VL + VC ≈ 0

This is one of the key indicators of resonance in a series RLC circuit. Even when VC and VL are individually large, their algebraic sum approaches zero because one leads and the other lags the current by 90 degrees.

For the component values used in this experiment, the theoretical resonant frequency is
$f_0 = 1 / (2\pi\sqrt{LC})$
$= 1 / (2\pi\sqrt{((1 \times 10^{-6})(1 \times 10^{-9}))})$
≈ 5.03 MHz
Thus, the expected resonance is around 5 MHz.

Resonance Measurement

To determine the resonant frequency experimentally, the input frequency was varied while observing the relevant voltages on the oscilloscope. Resonance was identified by examining the relationship between the capacitor and inductor voltages. In a series RLC circuit, resonance occurs when the magnitudes of VC and VL become approximately equal and opposite in phase, causing their combined voltage to approach a minimum value.

Using the oscilloscope probes and the Math function, the capacitor voltage was obtained and compared with the inductor-related measurement. The frequency was adjusted until the response indicated resonance behavior. Based on the oscilloscope observation, resonance was found near 5.7 MHz, which is reasonably close to the theoretical value of approximately 5.03 MHz.

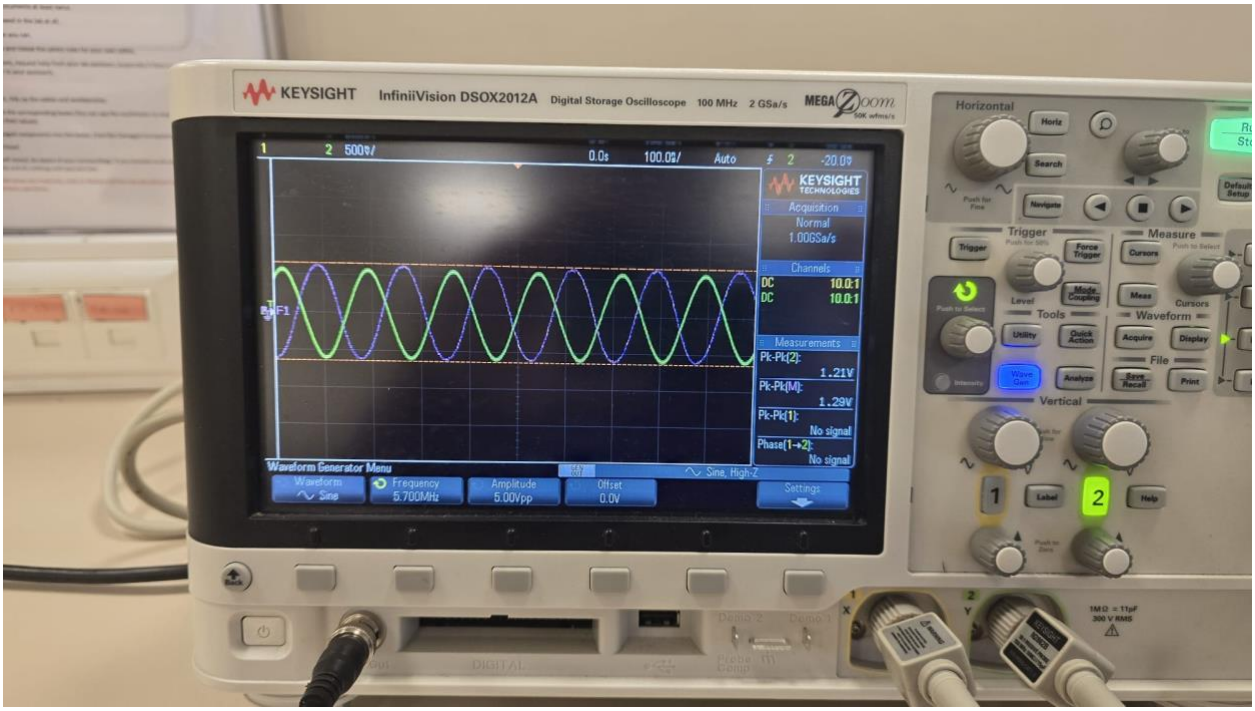


Figure 9. Oscilloscope display used to identify the resonant frequency of the series RLC circuit.

The difference between the theoretical and measured resonant frequencies may be due to practical non-idealities such as component tolerances, parasitic resistance and capacitance of the breadboard, internal resistance of the inductor, and limitations introduced by the measurement setup at high frequency. Since the experiment operates in the MHz range, even small stray capacitances or inductances from wires, probes, and breadboard connections can noticeably shift the resonant frequency.

Observation

The oscilloscope measurement taken near resonance shows that the relevant voltage signals become close in magnitude, supporting the expected resonance condition. This confirms that the circuit transitions from capacitive behavior to inductive behavior around the measured resonant frequency. Therefore, the measured response is consistent with the theoretical frequency response of a series RLC resonant circuit.

Part 2.d – Measurement of |VC + VL|, |VL|, and |VC|

In this part of the experiment, a 5 Vpp sinusoidal signal was applied to the series RLC circuit, and the frequency was adjusted to several points around the resonant frequency. The purpose of this step was to observe how the voltages across the capacitor and the inductor change as the operating frequency moves from below resonance to above resonance.

The measurements were taken at 0.1f0, 0.5f0, 0.8f0, f0, 1.5f0, 2f0, and 4f0. For each frequency, the magnitudes of |VC + VL|, |VL|, and |VC| were recorded. Since the capacitor was not directly referenced to ground, its voltage could not be measured directly with a single oscilloscope channel. For this reason,

the oscilloscope Math function was used. One channel measured the combined voltage across the capacitor and inductor, and the other channel measured the inductor voltage alone. Then the capacitor voltage was obtained as VC = (VC + VL) – VL. This method made it possible to determine the capacitor voltage indirectly and compare the behavior of all three voltages as a function of frequency.

Theoretical Background

In a series RLC circuit, the same current flows through the resistor, inductor, and capacitor. However, the voltage across each reactive element depends on its reactance, which varies with frequency. The inductive reactance is XL = ωL and the capacitive reactance is XC = 1 / ωC

At low frequencies, XC is very large while XL is small. Therefore, the capacitor dominates the circuit behavior, and a large portion of the reactive voltage appears across the capacitor. This is why, at frequencies well below resonance, |VC| is expected to be large and |VL| is expected to be relatively small.

As the frequency approaches the resonant frequency, XC decreases and XL increases. At resonance, the condition XL = XC is satisfied. Under this condition, the inductor and capacitor voltages become approximately equal in magnitude but opposite in phase. Because they are 180 degrees out of phase with respect to each other, they cancel when summed. Therefore, near resonance, the quantity |VC + VL| is expected to reach a minimum value even though |VC| and |VL| themselves may still be significant.

Above resonance, XL becomes larger than XC, so the circuit behaves more inductively. In this region, the voltage across the inductor becomes dominant, while the capacitor voltage decreases.

Measured Results and Interpretation

Figure 10.

500Khz 2500Khz 4MHz 5MHz 7.5MHz 10MHz 20MHz

	0.1f_0	0.5f_0	0.8f_0	f_0	1.5f_0	2f_0	4f_0
$\lvert V_C+V_L \rvert$	4.9V	2.13V	1.18V	760mV	860mV	1.49 V	3.19 V
$\lvert V_L \rvert$	56mV	500mV	820mV	1.07V	1.47 V	1.87 V	3.27V
$\lvert V_C \rvert$	4.9V	2.5V	1.7 V	1.4V	900 mV	800mV	500mV

The experimental results follow the expected general trend of a series RLC circuit. At 0.1f0, the measured capacitor voltage was approximately 4.9 V, while the inductor voltage was only about 56 mV. This agrees

with theory, since at very low frequencies the capacitive reactance is much larger than the inductive reactance, causing most of the voltage to appear across the capacitor.

At 0.5f0 and 0.8f0, the capacitor voltage decreased to approximately 2.5 V and 1.7 V, respectively, while the inductor voltage increased to about 500 mV and 820 mV. This indicates that the circuit was moving toward resonance, where the reactive contributions of the capacitor and inductor become more balanced.

At the resonant frequency, the measured values were approximately

|VC + VL| = 700 mV
|VL| = 1.07 V
|VC| = 1.4 V

The important observation here is that |VC + VL| became much smaller than the individual reactive voltages. This is consistent with the theory of resonance, since the inductor and capacitor voltages oppose each other and partially cancel when added together. Ideally, at resonance, |VC| and |VL| should be very close to each other, and their sum should approach zero. In the experiment, the measured values were not exactly equal, but they were sufficiently close to show the expected resonance behavior.

Beyond resonance, the inductor voltage continued to increase while the capacitor voltage decreased. For example, at 1.5f0, |VL| was approximately 1.47 V while |VC| was about 900 mV. At 2f0, the inductor voltage increased further to 1.87 V, while the capacitor voltage decreased to 800 mV. At 4f0, the inductor voltage reached about 3.27 V, while the capacitor voltage dropped to about 500 mV. These results are consistent with the expected high-frequency inductive behavior of the circuit.

It is important to note that some of the measured values did not match the ideal theoretical expectation very closely. In particular, near resonance, |VC| and |VL| were expected to be closer to one another. However, there were practical difficulties during measurement, especially related to the oscilloscope probes. Because of these probe-related issues, some measurements could not be obtained as accurately as desired, and this likely affected the closeness of the recorded values. In addition, at MHz frequencies, small non-ideal effects such as probe loading, parasitic capacitance and inductance, breadboard effects, and connection length can significantly influence the measured voltages. Therefore, although the numerical values were not perfectly matched, the overall experimental trend still agrees with the theoretical response of a series RLC resonant circuit.

Overall, the measurements confirm the key theoretical behaviors of the circuit: the capacitor voltage dominates below resonance, the inductor voltage dominates above resonance, and the combined reactive voltage |VC + VL| becomes minimum near resonance due to cancellation between VC and VL.

Part 2.e – Lower and Upper 3-dB Frequencies, Bandwidth, and Quality Factor

In this part of the experiment, the lower and upper 3-dB frequencies of the series RLC circuit were determined, and the bandwidth and quality factor were calculated. These quantities are important because they describe how sharply the circuit resonates and how selective it is around the resonant frequency.

Theoretical Background

A series RLC circuit consists of a resistor (R), an inductor (L), and a capacitor (C) connected in series. The behavior of this circuit strongly depends on the frequency of the applied input signal. This is because the inductor and capacitor store energy in different forms and respond differently to changes in frequency. The inductor stores energy in a magnetic field, while the capacitor stores energy in an electric field. As the frequency of the input signal changes, the opposition of these elements to current flow, known as reactance, also changes.

The inductive reactance is given by $X_L = \omega L$ and the capacitive reactance is $X_C = 1 / (\omega C)$ where $\omega = 2\pi f$ is the angular frequency of the input signal. As the frequency increases, the inductive reactance increases linearly, while the capacitive reactance decreases. Because of this opposite behavior, there exists a specific frequency at which the effects of the inductor and capacitor cancel each other. This condition occurs when $X_L = X_C$ or equivalently $\omega L = 1 / (\omega C)$

At this point, the circuit reaches resonance. The resonant frequency of a series RLC circuit is therefore given by $f_o = 1 / (2\pi\sqrt{LC})$

At the resonant frequency, the inductive and capacitive reactances are equal in magnitude but opposite in phase. As a result, they cancel each other, and the circuit behaves as if it only contains the resistor. This means the total impedance of the circuit becomes minimum and purely resistive. Consequently, the current in the circuit reaches its maximum value for a given input voltage.

Using the component values used in the experiment, L = 1 µH and C = 1 nF, the theoretical resonant frequency can be calculated as $f_o \approx 5.03$ MHz

Another important concept in resonant circuits is the 3-dB frequencies. These frequencies correspond to the points where the power of the circuit decreases to half of its maximum value. In terms of voltage magnitude, this corresponds to a decrease by a factor of $1/\sqrt{2}$ relative to the reference magnitude. These two frequencies are called the lower cutoff frequency (fL) and the upper cutoff frequency (fH). They define the frequency range in which the circuit effectively operates around resonance.

The difference between these two frequencies is known as the bandwidth of the circuit $BW = fH - fL$

Bandwidth describes how wide the frequency region is in which the circuit responds strongly. For a series RLC circuit, the theoretical bandwidth can also be expressed as $BW = R / (2\pi L)$

Using the resistance value $R = 40\ \Omega$ and the inductance $L = 1\ \mu H$, the theoretical bandwidth becomes $BW \approx 6.366$ MHz

Another important parameter that characterizes resonant circuits is the quality factor (Q). The quality factor measures how sharp or selective the resonance is. It indicates how efficiently the circuit stores energy relative to how quickly the energy is dissipated by the resistor. The quality factor is defined as $Q = f_o / BW$

Using the calculated values, $Q = 5.03 / 6.366$, $Q \approx 0.79$

A smaller Q value indicates that the resonance peak is relatively broad and that the circuit responds to a wider range of frequencies around the resonant frequency. In contrast, a higher Q value would indicate a very sharp resonance where the circuit responds strongly only to frequencies very close to f_o.

Therefore, in this circuit the relatively low quality factor suggests that the resonance is not very sharp and that the circuit has a relatively wide operating frequency range around the resonant frequency.

Experimental Evaluation

In the experiment, the values of fL and fH were first determined from the theoretical calculations written on paper, and then the function generator and oscilloscope were used to examine the circuit response near those frequencies. Based on these calculations, the lower and upper cutoff frequencies were taken as approximately

$fL \approx 2.77$ MHz
$fH \approx 9.14$ MHz

These values were then checked experimentally. Due to practical measurement limitations, the measured frequencies were not expected to match the theoretical values exactly. However, frequencies close to these calculated values were observed during the experiment, showing that the theoretical analysis and the experimental response were in reasonable agreement.

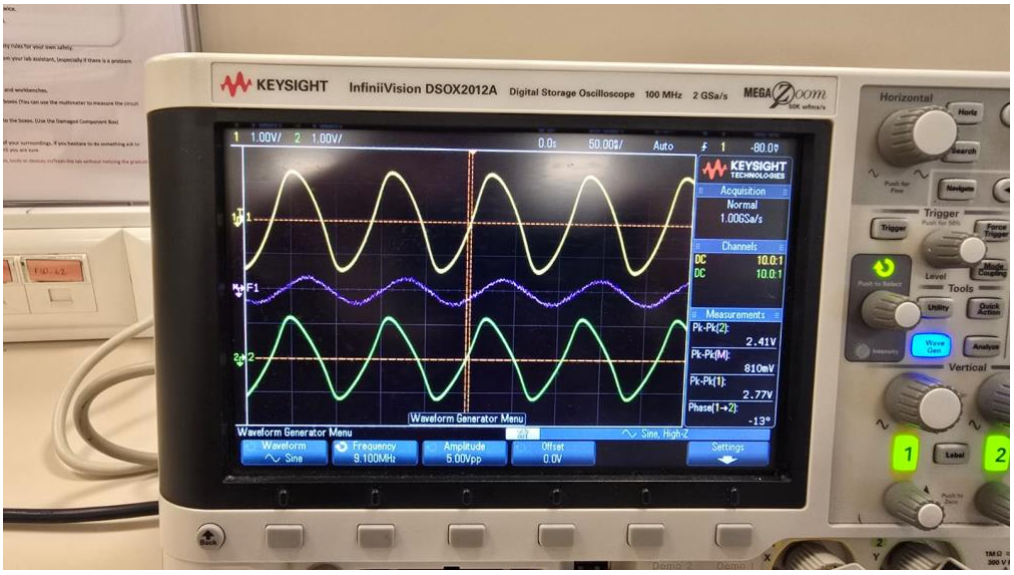


Figure 11. Oscilloscope measurement of the series RLC circuit response at a frequency near the upper cutoff region, showing the measured waveforms of the signals used to determine V_L, V_C, and $V_C + V_L$.

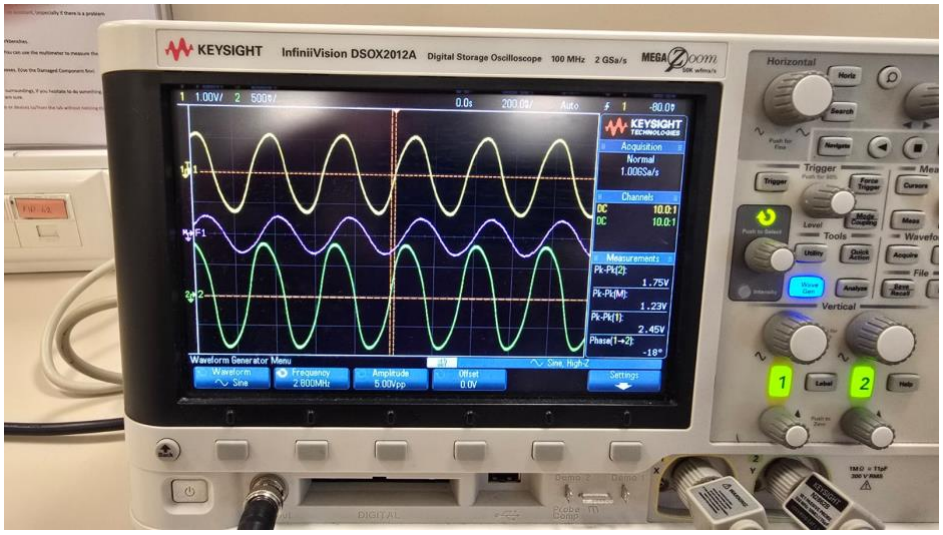


Figure 12. Oscilloscope measurement of the series RLC circuit at a lower frequency region, illustrating the relative magnitudes and phase relationship between the measured signals.

This agreement is important because it confirms that the circuit behaved close to the expected resonant model. The small differences between theoretical and measured values can be explained by several practical factors, including component tolerances, non-ideal characteristics of the inductor and capacitor, parasitic effects of the breadboard and wires, and probe-related loading at high frequencies. Since the experiment was performed in the MHz range, even small parasitic effects could shift the cutoff frequencies and affect the measured amplitudes.

The lower cutoff frequency represents the transition from the strongly capacitive region toward the resonant region, while the upper cutoff frequency marks the transition from the resonant region toward the inductive region. Together, these two frequencies define the bandwidth of the circuit. Since the calculated bandwidth is relatively large compared to the resonant frequency, the quality factor is less than 1, which indicates that the resonance is broad rather than sharply peaked.

Even though the experimentally observed cutoff points were not exactly identical to the calculated values, the measured response was close enough to support the theoretical predictions. Therefore, the experiment successfully demonstrated the concepts of lower and upper 3-dB frequencies, bandwidth, and quality factor in a series RLC circuit.

4. Discussion/Analysis

The experimental results obtained from the series RLC circuit generally follow the expected theoretical behavior of a resonant circuit. According to theory, the circuit should exhibit resonance near the calculated resonant frequency of approximately 5.03 MHz, where the inductive and capacitive reactances become equal. At this point, the voltages across the inductor and capacitor are expected to be approximately equal in magnitude but opposite in phase, causing the combined voltage $|V_C + V_L|$ to become minimal.

The measurements collected at different normalized frequencies confirm this behavior. At very low frequencies such as $0.1f_0$, the capacitor voltage dominates the circuit response. In the experiment, the measured capacitor voltage was approximately 4.9 V while the inductor voltage was only about 56 mV. This is consistent with theory because at low frequencies the capacitive reactance X_C is very large while the inductive reactance X_L is small, causing most of the voltage to drop across the capacitor.

As the frequency increased toward the resonant frequency, the capacitor voltage gradually decreased while the inductor voltage increased. For example, at $0.5f_0$ the capacitor voltage dropped to about 2.5 V while the inductor voltage increased to around 500 mV. At $0.8f_0$, the capacitor voltage further decreased to approximately 1.7 V while the inductor voltage increased to about 820 mV. This trend reflects the expected transition toward resonance, where the reactive effects of the capacitor and inductor become more balanced.

Near the resonant frequency f_0, the measurements show that the magnitudes of V_L and V_C become closer to each other. In the experiment, the measured values were approximately $|V_L| \approx 1.07\,V$ and $|V_C| \approx 1.4\,V$. At the same time, the magnitude of $|V_C + V_L|$ decreased to about 700 mV. This behavior is characteristic of resonance in a series RLC circuit, since the voltages across the inductor and capacitor are out of phase and partially cancel each other when summed.

Beyond the resonant frequency, the circuit begins to behave more inductively. As expected, the inductor voltage continued to increase while the capacitor voltage decreased. For instance, at $1.5f_0$ the inductor voltage increased to approximately 1.47 V while the capacitor voltage decreased to about 900 mV. At higher frequencies such as $2f_0$ and $4f_0$, the inductor voltage became dominant, reaching values of approximately 1.87 V and 3.27 V respectively, while the capacitor voltage continued to decrease. This confirms the theoretical expectation that inductive reactance increases with frequency while capacitive reactance decreases.

Although the overall trends observed in the measurements match the theoretical predictions, the values were not perfectly symmetric around resonance. Ideally, the magnitudes of V_C and V_L should be nearly equal at resonance, and their sum should approach zero. However, in the experiment the values were not exactly equal and the minimum value of $|V_C + V_L|$ was not exactly zero. One possible reason for this discrepancy is the difficulty in obtaining precise measurements at high frequencies using the oscilloscope probes. During the experiment there were some issues related to probe positioning and measurement stability, which may have affected the accuracy of some readings.

In addition, practical non-ideal effects such as component tolerances, parasitic capacitances and inductances of the breadboard and connecting wires, and probe loading can influence measurements at MHz frequencies. These factors can slightly shift the effective resonant frequency and alter the measured voltage magnitudes. Despite these small discrepancies, the experimental results still clearly demonstrate the fundamental behavior of a series RLC circuit, including capacitive dominance at low frequencies, voltage cancellation near resonance, and inductive dominance at higher frequencies.

The measurements, theoretical calculations, and observed oscilloscope waveforms are in reasonable agreement and successfully illustrate the frequency response characteristics of the series RLC resonant circuit.

5. Post-Lab

Question 1: What are the key differences in the frequency response of series RC and RL circuits?

Answer:
The key difference between the frequency response of series RC and series RL circuits comes from the opposite frequency dependence of capacitors and inductors. In a series RC circuit, the capacitive reactance decreases as frequency increases. At low frequencies, the capacitor has a large reactance, so the output voltage across the capacitor remains relatively large. As the frequency increases, the capacitor reactance decreases and the voltage across the capacitor drops. For this reason, when the output is taken across the capacitor, the series RC circuit behaves as a low-pass filter. It allows low-frequency components to appear strongly at the output while attenuating high-frequency components.

In contrast, in a series RL circuit, the inductive reactance increases with frequency. At low frequencies, the inductor behaves almost like a short circuit because its reactance is small, while at high frequencies

its reactance becomes large. Therefore, if the output is taken across the inductor, the voltage across it becomes larger as frequency increases, and the circuit behaves as a high-pass filter. This means low-frequency components are weaker at the output, while high-frequency components are emphasized.

Another important difference is the phase behavior. In the RC circuit, the output across the capacitor lags the input, and the phase shift becomes more significant as frequency increases. In the RL circuit, the output across the inductor leads the resistor voltage behavior differently because the inductor stores energy in a magnetic field and opposes changes in current. Thus, although both circuits are first-order frequency-dependent circuits, their magnitude and phase responses vary in opposite directions because capacitors and inductors respond to frequency in fundamentally different ways.

Question 2: In a series resonant circuit, which component dominates the equivalent impedance at very low and very high frequencies compared to f0?

Answer:
In a series resonant RLC circuit, the equivalent impedance depends on the relative sizes of the inductive reactance and capacitive reactance. At frequencies much lower than the resonant frequency f0, the capacitive reactance is very large, while the inductive reactance is small. As a result, the circuit behaves mainly as a capacitive circuit, and the capacitor dominates the equivalent impedance. In this region, most of the reactive opposition to current comes from the capacitor.

At frequencies much higher than f0, the situation is reversed. The inductive reactance becomes very large, while the capacitive reactance becomes small. Therefore, the circuit behaves mainly as an inductive circuit, and the inductor dominates the equivalent impedance.

At the resonant frequency itself, the inductive and capacitive reactances become equal in magnitude and opposite in sign. They cancel each other, so the total impedance becomes purely resistive and is determined mainly by the resistor. This is why the current becomes maximum at resonance in a series RLC circuit.

Question 3: Are the measured values of VC + VL in Figure 2 at any resonance different from your pre-lab calculations? If so, what might be the cause of this difference?

Answer:
Yes, the measured values of VC + VL near resonance were different from the ideal values predicted in the pre-lab calculations. According to theory, at resonance the voltages across the capacitor and inductor should be approximately equal in magnitude and opposite in phase, so their sum should become very small and ideally approach zero. In the experiment, VC + VL did decrease significantly near resonance, which confirms the expected resonance behavior, but it did not become exactly zero.

There are several reasons for this difference. First, real circuit components are not ideal. The resistor, inductor, and capacitor all have tolerances, meaning their actual values may differ slightly from their nominal values. The inductor also has internal resistance, and both the inductor and capacitor may introduce parasitic effects that are not included in ideal calculations.

Second, the experiment was performed in the MHz range, where parasitic capacitances and inductances from the breadboard, jumper wires, and probe connections become much more significant. These small non-ideal effects can shift the resonant frequency and change the measured voltages.

Third, during the measurements there were some practical difficulties related to the oscilloscope probes. Probe loading, imperfect positioning, and instability in the measurement setup can affect the measured values, especially when trying to compare two voltages that should nearly cancel each other. Therefore, even though the experimental value of VC + VL was not identical to the ideal pre-lab result, the observed trend still agreed with the theory and clearly demonstrated resonance.

Question 4: In Part 2, why cannot we connect the oscilloscope probe between the capacitor terminals directly to measure VC?

Answer:
In Part 2, the voltage across the capacitor cannot be measured directly by simply placing the oscilloscope probe across its terminals because of the grounding structure of a standard oscilloscope. In most laboratory oscilloscopes, the ground clips of all channels are internally connected to the same earth ground. This means that if the probe ground is connected to a point in the circuit that is not actually the circuit ground, it can unintentionally short that point to ground and disturb the circuit operation.

In the series RLC circuit used in this experiment, the capacitor is not directly connected between a node and the common ground reference. Instead, it is located between two floating nodes in the series branch. If the oscilloscope ground clip were connected directly across one terminal of the capacitor, it could force that node to ground and change the behavior of the circuit, or even create an unintended short circuit.

For this reason, the capacitor voltage was measured indirectly using two oscilloscope channels and the Math function. One channel measured the voltage across the capacitor and inductor together, while the second channel measured the voltage across the inductor alone. Then the capacitor voltage was obtained from the difference:
$VC = (VC + VL) - VL$

This method allows the capacitor voltage to be determined without disturbing the circuit by improper grounding.

Question 5: What is the working principle of a Network Analyzer? Discuss how it measures the frequency response of a circuit, including the roles of reference and measured signals, sweep types, and the importance of phase and magnitude analysis.

Answer:
A Network Analyzer is an instrument used to characterize how a circuit or system responds to signals over a range of frequencies. Its main purpose is to measure the frequency response of a device by comparing an input signal with an output signal. In general, the instrument applies a known sinusoidal test signal to the circuit and then measures how the circuit changes the magnitude and phase of that signal as frequency varies.

The working principle is based on comparing a reference signal and a measured signal. The reference signal represents the known input applied to the circuit, while the measured signal is taken from the output or another point of interest in the circuit. By computing the ratio of the measured signal to the reference signal, the analyzer determines the gain and phase shift of the circuit at each frequency. In this way, the instrument directly obtains the transfer function of the system in terms of magnitude and phase.

A network analyzer usually performs this measurement by sweeping the frequency of the input signal over a selected range. Different sweep types can be used depending on the application. In a linear sweep, the frequency changes in equal steps, which is useful when a uniform spacing of frequencies is needed. In a logarithmic sweep, the frequency changes by constant ratios, which is especially useful when the circuit behavior spans several decades of frequency, as in Bode plots. Some instruments may also support segmented or custom sweeps for more detailed analysis in specific frequency regions.

Magnitude analysis is important because it shows how much the circuit amplifies or attenuates signals at different frequencies. This helps identify behaviors such as low-pass, high-pass, band-pass, or resonant response. Phase analysis is equally important because it shows the time relationship between the input and output signals. In many circuits, especially those containing capacitors and inductors, phase information is essential for understanding energy storage, reactive behavior, resonance, and stability.

Together, magnitude and phase provide a complete picture of the circuit response. For example, in this lab the magnitude response helped identify attenuation regions and resonance behavior, while the phase response helped confirm the transition between capacitive and inductive dominance. Therefore, a network analyzer is a very useful tool because it automates and improves the measurement of frequency response more accurately and efficiently than manual measurements with a function generator and oscilloscope.

6. Conclusion

In this experiment, the frequency response characteristics of RC and RLC circuits were investigated through theoretical analysis, pre-lab calculations, and experimental measurements. The main objective was to understand how circuit behavior changes with frequency and to observe the resonance phenomenon in a series RLC circuit. By performing calculations and measurements at different frequencies, the relationship between capacitive and inductive reactances and their effect on circuit impedance was clearly demonstrated.

The experimental results were generally consistent with the theoretical expectations. At frequencies much lower than the resonant frequency, the circuit exhibited predominantly capacitive behavior, where the capacitor voltage was significantly larger than the inductor voltage. As the frequency increased toward the resonant frequency, the voltages across the capacitor and inductor became closer in magnitude, and the combined voltage $| V_C + V_L |$ approached a minimum value. This observation supports the theoretical prediction that at resonance the inductive and capacitive reactances cancel each other, leaving the circuit impedance primarily determined by the resistor.

Beyond the resonant frequency, the circuit behavior became more inductive, which was confirmed by the increase in the inductor voltage and the decrease in the capacitor voltage. These results demonstrate the expected transition of the circuit from capacitive dominance at low frequencies to inductive dominance at higher frequencies. The measured values therefore validated the theoretical model of the series RLC circuit and confirmed the predicted resonant behavior.

Although the general trends of the measurements matched the theoretical analysis, small discrepancies were observed between the calculated and measured values. These differences can be attributed to several practical factors such as component tolerances, parasitic capacitances and inductances of the breadboard and connecting wires, and the internal resistance of the inductor. In addition, measurement limitations and oscilloscope probe effects may have introduced small errors, especially when measuring voltages that nearly cancel each other near resonance.

Overall, the experiment successfully demonstrated the fundamental concepts of resonance, bandwidth, and frequency-dependent circuit behavior. By comparing theoretical calculations, expected results, and experimental observations, the lab reinforced the understanding of how reactive components influence circuit impedance and voltage distribution across different frequency ranges. These insights are essential for analyzing and designing circuits that rely on frequency response, such as filters and resonant systems.